

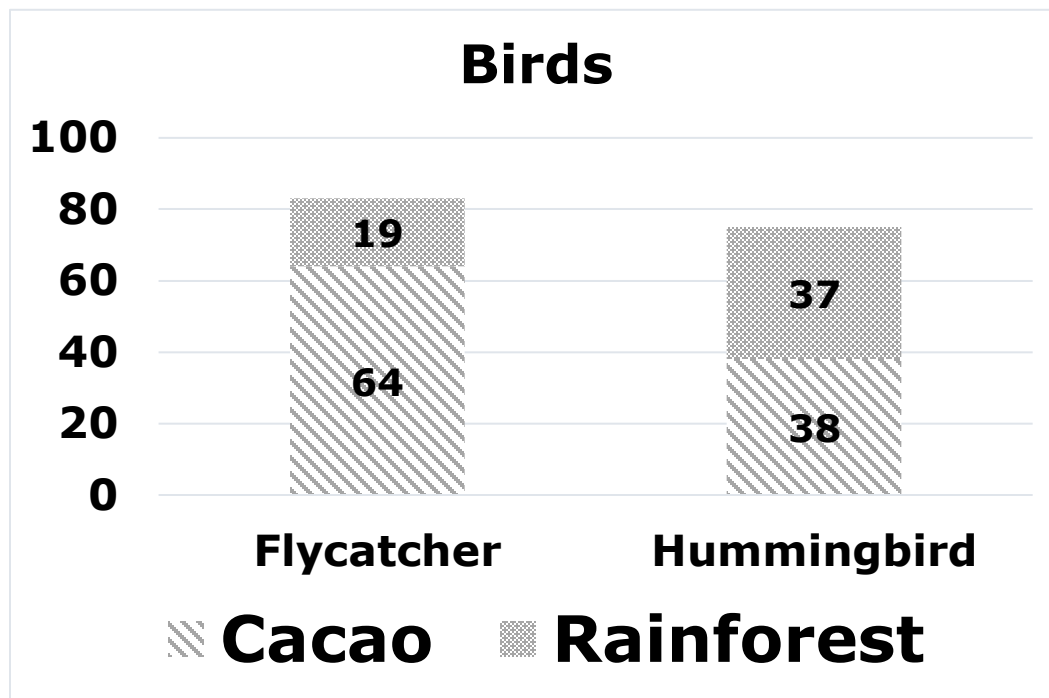
1. A survey asked 775 adults the question “Do you like it or dislike it when strangers start conversations with you?”.
- 48% participants identified as male.
 - 35% participants who identified as male stated that they like it when strangers start conversations.
 - 42% participants who identified as female stated that they like it when strangers start conversations.

Use the information to complete the two-way frequency table.

**Do you like it or dislike it when
strangers start conversations with you?**

	Like It	Dislike It	Total
Male	130	212	372
Female	169	234	403
Total	299	476	775

2. Skye is a biologist who conducted a study on the abundance of birds on cacao farms in Costa Rica compared to the abundance of birds in a nearby rainforest. Skye used a segmented bar graph, as shown, to display the data for two different species of birds.



- a. Use the segmented bar graph to complete the two-way relative frequency table.

	Flycatcher	Hummingbird	Total
Cacao Farm	40.51%	24.05%	64.56%
Rainforest	12.03%	23.42%	35.44%
Total	52.53%	47.47%	100.00%

- b. Interpret the joint relative frequency for Flycatchers found on a Cacao Farm.

40.51% of birds found were Flycatchers on the Cacao Farm

- c. Interpret the marginal relative frequency for birds found in the rainforest.

35.44% of birds were found in the rainforest

3. Students in each class at Leesburg High School were asked whether they have to do chores at home and whether they receive an allowance. Two-way frequency tables that show some of the results of the survey are shown.
- a. Complete each two-way frequency table.

Freshmen				Sophomores			
	Do Get An Allowance	Do Not Get An Allowance	Total		Do Get An Allowance	Do Not Get An Allowance	Total
Do Chores	96	25	121	Do Chores	131	62	193
Do Not Do Chores	60	121	181	Do Not Do Chores	18	58	76
Total	156	146	302	Total	149	120	269

Juniors				Seniors			
	Do Get An Allowance	Do Not Get An Allowance	Total		Do Get An Allowance	Do Not Get An Allowance	Total
Do Chores	101	72	173	Do Chores	105	84	189
Do Not Do Chores	71	23	94	Do Not Do Chores	21	42	63
Total	172	95	267	Total	126	126	252

- b. Determine the row conditional relative frequency for Sophomores who get an allowance and who do not do chores.

23.68%

- c. Determine the column conditional relative frequency for Freshman who do not get an allowance and who do not do chores.

82.88%

4. A survey found that of 2,000 high school seniors surveyed 1,800 have their driver's license. Of those who have their driver's license, 1,350 have a job. Of the seniors who do not have a driver's license, 40 have a job.

a. Complete a two-way relative frequency table.

	Has a Job	Does Not Have a Job	Total
Has Driver's License	67.5%	22.5%	90%
Does Not Have a Driver's License	2%	8%	10%
Total	69.5%	30.5%	100%

- b. Explain how the information in the table can be used to show that there is an association between having a job and having a driver's license.

About $\frac{3}{4}$ or the seniors surveyed that have a driver's license have a job, whereas only about $\frac{1}{5}$ of those who don't have a license have a job. This shows that there is an association between having a driver's license and having a job.